

Yonghao Tan

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Research Experience

5nm UCle-Enabled Multi-Chiplet Generalizable Rendering Processor

Mar. 2024 - Sept. 2025

AI Chip Center for Emerging Smart Systems, Hong Kong, China

- Architected a 5nm 4-chiplet GeNeRF processor for generalizable rendering to address the heavy external-memory traffic induced by multi-view feature fetching.
- Introduced a UCle-enabled cross-die unified cache, distributed source-view management, and balance-aware scheduling to maximize source-view reuse while reducing off-chip and cross-die data movement.
- Silicon results reached 91.43 TOPS/W, 55.43 FPS real-time rendering, and 0.29 uJ/pixel through hierarchical sparsity and hybrid NeRF-SR execution in a 45 mm x 45 mm MCM package footprint.

55nm ReRAM-on-Logic Stacked LLM Accelerator for Speculative Decoding

Apr. 2024 - Aug. 2025

AI Chip Center for Emerging Smart Systems, Hong Kong, China

- Architected a 55nm edge LLM accelerator whose logic die is stacked with four ReRAM dies via face-to-face bump bonding to support in-stack storage of draft-model codebooks.
- Developed block-clustered weight compression, local-rotation-based outlier-free W4A8 quantization, and adaptive parallel speculative decoding to reduce both target-model EMA and rejected-draft overhead.
- The prototype delivered 14.08 to 135.69 token/s and a 4.46x to 7.17x throughput speedup over a BF16 speculative-decoding baseline on a 55.98 mm² logic die.

28nm CNN-Transformer Accelerator for Semantic Segmentation

Nov. 2021 - Sept. 2024

AI Chip Center for Emerging Smart Systems, Hong Kong, China

- Architected a 28nm memory-compute-intensity-aware accelerator for high-resolution ConvFormer and SegFormer semantic-segmentation workloads.
- Combined hybrid-attention processing, data-reuse-oriented layer fusion, and cascaded feature-map pruning to reduce attention-side EMA and eliminate redundant KV and weight movement across fused blocks.
- Silicon results achieved 0.22 uJ/token on SegFormer-B0 and up to 52.90 TOPS/W peak efficiency in a 13.93 mm² chip.

Education

Ph.D. in Electronic and Computer Engineering

Sept. 2023 - Present

The Hong Kong University of Science and Technology (HKUST)

Hong Kong, China

Supervisor: Prof. Tim Kwang-Ting Cheng

B.E. in Microelectronics

Sept. 2019 - Jun. 2023

Southern University of Science and Technology (SUSTech)

Shenzhen, Guangdong, China

Supervisor: Prof. Fengwei An

Overall GPA: 3.77 / 4.0 | Weighted Average: 90.38 | Rank: 11 / 77

Publications

A 5nm 91.43 TOPS/W 4-Chiplet Generalizable-Rendering-Processor with UCle-Enabled Cross-Die-Cache and Balance-Aware Progressive Multi-Level Sparsity

Tan, Y.*, Ma, S.*, Dong, P., Luo, P., Lei, Z., Lu, W., Ying, G., ... & Cheng, K. T.
2026 IEEE Custom Integrated Circuits Conference (CICC), IEEE.

A 14.08-to-135.69Token/s ReRAM-on-Logic Stacked Outlier-Free Large-Language-Model Accelerator with Block-Clustered Weight-Compression and Adaptive Parallel-Speculative-Decoding

Dong, P., Tan, Y., Liu, X., Luo, P., Liu, Y., Pang, D., Ma, S., ... & Cheng, K. T.
2026 IEEE International Solid-State Circuits Conference (ISSCC), IEEE.

A 28nm 0.22uJ/Token Memory-Compute-Intensity-Aware CNN-Transformer Accelerator with Hybrid-Attention-Based Layer-Fusion and Cascaded Pruning for Semantic-Segmentation

Dong, P.*, Tan, Y.*, Liu, X., Luo, P., Liu, Y., Liang, L., ... & Cheng, K. T.
2025 IEEE International Solid-State Circuits Conference (ISSCC), IEEE.

Genetic Quantization-Aware Approximation for Non-Linear Operations in Transformers

Dong, P.*, Tan, Y.*, Zhang, D., Ni, T., Liu, X., Liu, Y., ... & Cheng, K. T.
2024 61st ACM/IEEE Design Automation Conference (DAC), IEEE.

A Reconfigurable Coprocessor for Simultaneous Localization and Mapping Algorithms in FPGA

Tan, Y.*, Deng, H.*, Sun, M., Zhou, M., Chen, Y., Chen, L., ... & An, F.
IEEE Transactions on Circuits and Systems II: Express Briefs, 70(1), 286-290, 2022.

A Reconfigurable Visual-Inertial Odometry Accelerator with High Area and Energy Efficiency for Autonomous Mobile Robots

Tan, Y.*, Sun, M.*, Deng, H., Wu, H., Zhou, M., Chen, Y., ... & An, F.
Sensors, 22(19), 7669, 2022.

** Authors marked with an asterisk contributed equally to the corresponding work.*

Honors and Awards

Best Teaching Assistant Award, Department of Electronic and Computer Engineering, HKUST	Aug. 2025
Outstanding Graduate (School Level), SUSTech	May 2023
First-Class Outstanding Students Scholarship with the highest score	Sept. 2022
Undergraduate Innovation and Entrepreneurship Training Program	Apr. 2022
Shenzhen Longsys Electronics Company Award (Top 2% in the School of Microelectronics)	Dec. 2021
First Prize, 2021 National College Students FPGA Innovation Design Competition (Top 22 out of 1,341 teams)	Dec. 2021
First Prize, 2021 International Competition of Autonomous Running Robots (1st place out of 34 finalist teams)	Oct. 2021

Funding and Support

Postgraduate Studentship (PGS) Award in HKUST	Sept. 2023 - Present
Undergraduate Innovation and Entrepreneurship Training Programs (Provincial Level)	Apr. 2022
Guangdong College Students' Scientific and Technological Innovation (Provincial Level)	Jul. 2021

Skills

Research Interests: Software/Hardware Co-Design, Model Compression, 3D Processing

Programming Languages: C, C++, Java, Python, SystemVerilog, Verilog HDL, VHDL

Professional Software: AutoCAD, Cadence, Design Compiler, IC Compiler II, MATLAB, Multisim, Silvaco

Languages: English (fluent), Mandarin (native), Cantonese (native)